GHOST TRIALS

Evidence before confidence.

Autonomous failure discovery for executable physical-AI controllers, with a controlled agent-verification extension

Nebius x NVIDIA Global AI Hackathon · Physical AI Track
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Experimental evaluation platform — not a safety certification. Agent domain is local, not yet deployed.

System and evidence at a glance

Capability	Demonstrated status
Robot simulation	Real headless MuJoCo gantry-arm manipulation on CPU
Autonomous search	Bounded adaptive trials, reproduction, mutation, empirical boundary and regression
Evidence	Telemetry, recipe, seed, controller digest, simulator version and independently recomputed trace metrics
Local agent domain	Bundled executable reference agent, mock tools, versioned assertions, verified failure replay and case-specific regression; not deployed or open to arbitrary code
NVIDIA / Nebius	One completed live Nemotron 3 Super review of a verified MuJoCo trial through Token Factory
Not demonstrated	Physical robot, G1/SONIC/GR00T worker, live Cosmos generation, Nebius AI Cloud GPU or safety certification

Live application: https://ghosttrials.site/

Public source and MIT license: https://github.com/linoxbt/ghost-trials

Scope: an experimental policy-failure discovery platform, not a certified robot-safety assessment.

Why Ghost Trials exists and how it works

Inspiration

A robot controller may succeed under nominal conditions and fail when friction, object placement, mass or timing shifts. An AI agent can likewise complete a routine task but follow malicious tool output into an unauthorized request. In both cases, the useful question is not simply “What score did it get?” It is: **Which bounded condition caused the failure, what did the executable target actually do, and can a reviewer inspect and rerun the evidence?**

Ghost Trials grew from that question. We began with an experimental red-team workflow for embodied AI: give it an executable controller, define a task, and let it search *bounded, simulator-supported* conditions for weaknesses. We then applied the same evidence-first Hunt lifecycle to a tightly controlled AI-agent reference target. The product is not a certification authority, and passing a bounded set of trials does not establish general safety or security.

What it does

The shared workflow is **Select executable target → Define task and constraints → Release the Ghost → Inspect trials → Reproduce failure → Save Ghost Case → Compare another version → Hunt again**. The Physical AI controller task remains the submission's primary demonstration; the agent domain shows how the evidence workflow extends without pretending tool calls are robot telemetry.

The demonstrated default task runs entirely on CPU. A two-axis gantry arm pushes a free block across a tabletop toward a target band. MuJoCo advances articulated motion, gravity, contact and friction; the selected executable controller issues actions on each physics step. The default `gantry-push-v1` controller is a deliberately simple time-based ramp, and the bundled MLP example uses hand-set weights; they are test targets for the Hunt, not trained policies. The Hunt can vary object mass, surface friction, initial block position and controller delay—parameters that affect the implemented model. Unsupported controls are rejected rather than displayed as decorative settings.

The Hunt samples the declared search space, then uses observed failures and near-failures to focus later candidates. It mutates nearby physical conditions, repeats potential failures and, when a matched pass/fail pair exists, reports an **empirical boundary within the tested region**. It does not claim to find a global worst case or universal safety threshold. A verified finding becomes a Ghost Case: a reusable recipe, seed, controller digest, simulator version, telemetry and artifact manifest. Exact replay and compatible-controller regression answer the narrow question “Did this candidate pass this case?”—not “Is the candidate generally safe?”

The workspace presents Hunt, Failures, Trials and Compare views, with team, physical-recording and report tools under Account. It shows trial progress, measured task outcome, a telemetry-derived state replay, search observations, evidence checks, model interpretation and known limitations. The replay is generated from recorded simulator states; it is **not camera footage or a Cosmos-generated robot response**. Authenticated users have organization-isolated records. New workspaces do not automatically receive paid-inference access.

A second domain verifies a **bundled executable AI agent** in a controlled mock-tool environment. The bundled reference agents are small scripted programs (no LLM inside), chosen so each verdict is deterministic and reproducible; the harness treats them as opaque executables. A bounded Hunt changes user and untrusted tool-output inputs, records actual tool requests, and applies versioned deterministic assertions to unauthorized actions. It retains a trace and evidence digest, reproduces a saved failure, and tests a corrected reference agent against that case. The domain was demonstrated locally and its protected API is deployed on Railway, but the public Netlify UI has not yet been released with the necessary route. This is a deliberately narrow verification harness, not permission to execute arbitrary customer code on Railway. The agent domain is complementary to the Physical AI submission; it has not replaced the measured robot-physics path.

How we built it

The backend uses FastAPI, Pydantic and SQLite with a durable job queue, typed trial records, private artifact routes, role checks, budget admission and audit records. The frontend is a responsive multi-page application with Privy-backed sign-in. CPU simulation is a distinct adapter within the existing `ScenarioGenerator → Simulator → TrialJudge → evidence` pipeline; optional Cosmos, Isaac and SONIC/G1 integrations remain modular and fail closed when their resources are absent. Synthetic fixtures are development-only and cannot silently replace real execution.

Evidence is not a single LLM answer. For each completed CPU trial, the system retains the full physical recipe, random seed, controller identity/digest, simulator version, sampled state trajectory, measured metrics, task outcome, state-derived visualization and hashes. A separate verifier checks artifact integrity and recomputes selected results from the trace. Missing or corrupt evidence is not converted into a policy verdict. Provider outages are recorded as provider failures, not robot-policy failures. Observation and AI interpretation remain separate in the UI and API.

For the agent domain, the bundled harness executes the reference agent behind a parent-mediated mock-tool interface. It records attempted tool names and arguments, whether each request was authorized, state changes and assertion results. A denied forbidden request is reported as an **attempted unauthorized invocation**, not as a completed destructive action. Timeout or harness failure is an execution error, not a verified agent failure. The current process boundary is suitable for this bundled reference path only; it is not a security sandbox for arbitrary third-party executables.

NVIDIA models and Nebius

Ghost Trials uses **NVIDIA Nemotron 3 Super** (`nvidia/nemotron-3-super-120b-a12b`) through **Nebius Token Factory's runtime inference API** to interpret an artifact-verified CPU trial. The reviewer receives the measured task outcome, selected numeric metrics, recipe, simulator/checkpoint identities and evidence manifest. It must return structured JSON and cite exact supplied metric values. Invalid or invented citations are rejected; disagreement is flagged; a model failure cannot overwrite the deterministic MuJoCo result.

On 18 October 2026, an operator acceptance trial completed real MuJoCo simulation and a live Token Factory call. Trial `trl_9dad0149b31e` ran `gantry-push-v1` for **3.0 simulated seconds** with seed 42 and placed the block at **0.729056 m**, inside the **0.64–0.86 m** target band. The evidence checker reported `cpu_trace_metrics_verified` after checking **121 telemetry samples and four core artifacts**. Nemotron Super returned a completed **pass** interpretation, with **801 prompt tokens and 375 completion tokens**. This is one observed integration result, not a model-accuracy benchmark or independent safety review. An earlier attempt returned an invalid model response; the app retained the physical measurement and now records a safe error reason. The successful operator report was generated locally; the repository's public seed-42 reference bundle contains separately generated physics evidence **without** a model review. A fresh signed-in customer trial on the deployed site remains a separate validation step.

Token Factory removed the need to procure a GPU merely to host the judge. The robot physics still executes locally or on ordinary CPU infrastructure; remote model inference is not robot execution. The CPU reviewer makes at most one Super request per admitted verified trial and has no automatic Ultra fallback. The new agent reviewer is optional and budget-gated; its provider behavior has contract tests, **not a completed live Nemotron agent review**. The older optional GPU judge path contains two-pass Super/Ultra escalation, but that path has **not** been demonstrated with a G1 checkpoint. Application budget reservations are admission controls, not a guarantee about an external bill. The owner confirmed Nebius automatic top-up is off before the live acceptance call.

Challenges, learning and impact

The hardest boundary was resisting plausible but unsupported claims. A polished synthetic clip, a model's confident prose or an artifact hash can each look persuasive without proving that a policy ran. We made measured simulator state the source of physical observations, labeled synthetic and unauthenticated data, and displayed unavailable conclusions when evidence is insufficient. Hashes and metric recomputation detect several inconsistencies but do not prove trusted hardware execution, prevent every self-consistent forgery or establish transfer to a physical robot.

We also learned that reproducibility is more than a random seed. Controller digest, simulator version, recipe, timing, metric definitions and evidence provenance all matter. A useful failure report should make those identities visible so a robotics engineer can reproduce and challenge the finding.

The primary users for this Physical AI submission are robotics teams and policy developers who need to prioritize regressions before deploying a new checkpoint. The controlled agent milestone also addresses developers testing tool-use boundaries in executable agent programs. The immediate demonstrated value is a repeatable, evidence-backed adversarial test of an **executable controller in a defined simulator task** and of **bundled reference agents with mock tools**. Neither result is a regulator-approved certification, a general safety/security probability, or a replacement for physical validation and expert review.

What we accomplished and what remains

A bounded local adaptive CPU Hunt completed **16 real MuJoCo trials and found three failures** in the 18 October validation. A saved case replayed with the same measured final block position (**0.625826 m**); a different compatible controller executed the same recipe and reached **0.776821 m**, passing that *particular* case. In a separate 20 October two-domain rerun, a nine-trial MuJoCo grid Hunt found **three controller failures**. One saved case replayed at **0.499450 m**; the candidate controller reached **0.513131 m** and **still failed** that case, so the application reported no fix. Search outcomes depend on the sampled recipes; these runs have different strategies and must not be conflated.

The same demonstration ran a six-trial Hunt against a deliberately vulnerable bundled reference agent. Adversarial tool output induced a request for forbidden `admin.delete`; the mock-tool mediator denied the operation, and the recorded unauthorized request triggered a versioned assertion. The verified Ghost Case reproduced, while `reference-corrected-v2` passed that **specific** saved case. This is real local agent-program execution with mock tools, not a live customer-agent assessment or evidence of a production-service compromise. The full automated suite passed **193 tests** in the current development environment after the 20 October audit remediation. Tests include actual MuJoCo execution, bundled-agent execution and synthetic search fixtures; they are not independent robotics validation.

On 20 October, Railway deployment `2d0dd1f8-879e-4cd7-9448-6041089b834f` reached `SUCCESS`. The live `/health` response reported `multidomain-v1` and MuJoCo CPU execution; an unauthenticated direct `/agent-trials` request returned the expected `401` rather than `404`. These checks establish backend deployment and route protection, **not** a completed signed-in production Hunt. By 22 October the source was on `main` and the Netlify `/agent-trials` route returned the expected `401`, so the hosted frontend and backend match; a signed-in hosted two-domain Hunt is still the outstanding validation. Live Nemotron review is configured on the backend, but paid review still requires an operator-assigned budget for a particular signed-in organization; no public new account receives it automatically. The final Physical AI video should show at least one minute of actual Hunt, policy execution and evidence views, and should show a completed model review only if that signed-in trial really received one.

The original Unitree G1/GR00T/GEAR-SONIC/Isaac concept remains an optional, unvalidated integration path. No physical G1, compatible fine-tuned G1 checkpoint, end-to-end G1 worker run, live Cosmos generation or Nebius AI Cloud GPU instance is claimed. The next engineering milestones are a signed-in production Hunt with a budgeted review, an independent robotics assessment of trace and measurement limitations, and a compatible G1 run if hardware/compute access becomes available.

Evidence and reproducibility notes for judges

Public, inspectable CPU reference: index, telemetry, recipe, manifest, trajectory. This separately generated seed-42 run is **not** the model-reviewed acceptance trial. It is excluded from all customer results and scores.

Observed local validation: a 16-trial adaptive MuJoCo Hunt found three failures; a saved case replayed at 0.625826 m and a candidate controller reached 0.776821 m on the same case. A separate live-reviewed seed-42 trial reached 0.729056 m, passed trace verification and received the Nemotron Super review described above. The operator acceptance output was local, not a publicly archived raw evidence bundle. Independent human robotics review, video decoding and hardware attestation were **not** performed for that CPU trial. The trajectory SVG is derived from states, not video.

20 October two-domain local demonstration: `uv run python -m scripts.multidomain_demo --output data/multidomain-demo` executes nine real headless MuJoCo trials and six bundled-agent trials with no paid provider. The observed rerun had three controller failures; the selected case replayed at 0.499450 m, and the candidate still failed it at 0.513131 m. The agent run discovered and denied an unauthorized `admin.delete` request, verified and reproduced the trace, and found that the corrected reference agent passed that exact case. These are two different bounded test domains. The agent review integration was contract-tested, not called live in this demonstration. The selected physical case had no automatic reproduction attempt or empirical boundary in this particular nine-trial run; those are optional Hunt findings, not implied by exact replay.

Hosted release check (20 October): the Railway backend deployment above passed its platform verdict and returned the expected multi-domain health contract. The protected agent API returned `401` without a session. Re-checked on 22 October: the Netlify agent route returned `401` (proxied and protected) and public `main` holds the source. No signed-in hosted agent Hunt, production replay or hosted agent-model review was run as part of these checks. Regenerate the PDF dossier from this file before attaching it.

Run locally: follow the README, then run `uv run python -m scripts.cpu_smoke --strategy grid --trials 9 --output data/cpu-validation` for actual headless MuJoCo trials, exact replay and regression. Tests: `uv run --frozen --extra dev --extra physics --extra models --extra storage pytest -q`. The default CPU path needs no GPU or paid API. To add a live model review, use only the backend configuration and spending safeguards in the README; a customer's organization needs an explicit budget allocation.

Known limits: The default task is a gantry manipulation benchmark, not Unitree G1 locomotion. The built-in lightweight neural-policy format is executable on CPU but is not a trained G1 checkpoint. Physical recording support has only fixture-level validation. The agent harness accepts bundled reference agents only; it is not an arbitrary-code sandbox, and its corrected agent passed one specific regression case. No live Cosmos video, G1/SONIC/GR00T execution, Isaac worker acceptance, independent safety certification, or physical robot result is represented as complete. An application admission budget is not an external provider charge ceiling. Results are conditional on the tested controller or target, runtime version, recipe, seed, task and metric definition.